

Nutrition Fundamentals

Healthy Eating Principles

A healthy diet can take many forms. There is no single food pattern that every person must follow.

A useful framework is based on four principles:

1. **Adequacy** - consuming enough energy and nutrients to meet the body's needs.
2. **Balance** - balancing energy intake with expenditure and consuming appropriate amounts of protein, carbohydrate and fat.
3. **Moderation** - limiting foods or nutrients that can be harmful when consumed excessively.
4. **Diversity** - eating a wide variety of nutritious foods.

For most people, the foundation of the diet should include:

- vegetables;
- fruits;
- pulses and legumes;
- whole grains;
- nuts and seeds;
- dairy or suitable alternatives;
- eggs, fish, poultry or other lean protein sources when included in the diet.

Highly processed foods do not need to be completely avoided, but a diet dominated by foods high in sodium, free sugars, saturated fat or industrial trans fat is not ideal for long-term health.

Fruits and Vegetables

Fruits and vegetables provide:

- vitamins;
- minerals;
- fibre;
- water;
- plant compounds.

The World Health Organization recommends that people older than 10 years aim for at least about **400 g of fruits and vegetables per day**.

Fresh, frozen and appropriately canned fruits and vegetables can all contribute to a healthy diet.

Whole fruit is generally preferable to obtaining most fruit intake from juice because whole fruit retains more intact fibre and is usually more filling.

Dietary Fibre

Dietary fibre supports:

- bowel regularity;
- gut health;
- satiety;
- healthy blood glucose responses;
- healthy cholesterol levels.

Good sources include:

- vegetables;
- fruit;
- whole grains;
- lentils;
- beans;
- chickpeas;
- nuts;
- seeds.

For most people older than 10 years, WHO recommends at least about **25 g of naturally occurring dietary fibre per day**.

Fibre intake is usually best increased gradually if a person's usual intake is low.

Free Sugars

Free sugars include sugars:

- added during manufacturing or cooking;
- added by the consumer;
- present in honey;
- present in syrups;
- present in fruit juice;
- present in fruit-juice concentrates.

WHO recommends limiting free sugars to less than about **10% of total daily energy intake**.

Sugars naturally present inside intact fruits and milk are not treated in the same way as free sugars.

Sodium and Salt

Sodium is necessary for normal physiology, but excessive intake can contribute to high blood pressure.

WHO recommends that adults consume less than:

- **2,000 mg sodium per day**
- approximately **5 g of salt per day**

Common high-sodium foods include:

- packaged snacks;
- processed meats;
- instant noodles;
- sauces;
- pickles;
- restaurant foods;
- heavily processed foods.

Dietary Fat Quality

Dietary fat is necessary for:

- cell membranes;
- hormone production;
- absorption of fat-soluble vitamins;
- essential fatty acids;
- energy.

Unsaturated Fat Sources

Common sources include:

- nuts;
- seeds;
- peanuts;
- olive oil;
- mustard oil;
- other vegetable oils;
- avocado;
- fatty fish.

Saturated Fat

Common sources include:

- butter;
- ghee;
- high-fat dairy;
- fatty meats;
- coconut oil;
- palm oil;

- many bakery and processed foods.

WHO recommends keeping saturated fat below about **10% of total energy intake**.

Trans Fat

Industrially produced trans fat should be avoided as much as possible.

WHO recommends keeping total trans-fat intake below about **1% of total energy intake**.

Micronutrients

Micronutrients are vitamins and minerals required in relatively small amounts but are essential for health.

Iron

Iron contributes to oxygen transport and energy metabolism.

Food sources include:

- meat;
- fish;
- eggs;
- lentils;
- beans;
- chickpeas;
- leafy green vegetables;
- fortified foods.

Vitamin C consumed with plant-based iron sources can improve iron absorption.

Calcium

Calcium contributes to:

- bone health;
- muscle contraction;
- nerve function.

Sources include:

- milk;
- curd or yogurt;
- paneer;
- calcium-fortified plant milks;
- some tofu;
- leafy greens;
- sesame products.

Vitamin D

Vitamin D contributes to bone and muscle health.

Sources include:

- sunlight exposure;
- some foods;
- fortified foods;
- supplements when appropriate.

Vitamin B12

Vitamin B12 is naturally concentrated in animal-derived foods.

People following vegan diets need a reliable source of vitamin B12 through fortified foods or appropriate supplementation.

Magnesium

Magnesium participates in many biological processes, including muscle and nerve function.

Sources include:

- nuts;
- seeds;
- legumes;
- whole grains;
- leafy green vegetables.

Potassium

Potassium supports fluid balance, nerve function and muscle function.

Sources include:

- fruits;
- vegetables;
- potatoes;
- beans;
- lentils;
- dairy foods.

Food Labels

Nutrition labels can help compare foods.

Useful fields include:

- serving size;
- calories;

- protein;
- carbohydrate;
- sugars;
- added sugars where listed;
- fibre;
- total fat;
- saturated fat;
- sodium.

A food should not be judged from one nutrient alone. The overall nutrient profile matters.